

## **Financial Analytics Assignment 4 (02022024)**

### **Question 1**

The erection of the Hospital Building is under consideration.

There are two technical acceptable alternatives:

A reinforced concrete shell roof structure having an initial cost of **2,700,000** and a steel-framed structure with brick cladding for an initial cost of **1,800,000**. The life of the concrete building is estimated to be **60 years** and, while there will be no maintenance costs for the building during the first **10 years**, there will thereafter be an annual maintenance cost of **35,000**.

The life of the other building is estimated to be **20 years** with an equivalent annual maintenance cost from completion of construction of **40,000**. The salvage value of the concrete building is estimated at **80,000** and that of the steel-framed building at **27,000**.

An acceptable rate of return is assessed at **9.65 %**.

**Which is the better economic proposition? Make the comparisons based on:**

1. The equivalent annual cost method
2. The present value method.

### **Question 2**

A project has a first cost of Rs.10,000, net annual benefits of Rs.2,000, and a salvage value of Rs.3,000 at the end of its 10-year useful life. The project will be replaced identically at the end of 10 years, and again at the end of 20 years.

**What is the present worth of the entire 30 years of service if the interest rate is 11.43%?**

### **Question 3**

A Military project requires an initial investment of Rs.10 million and a supplementary investment of Rs.5 million at the end of the tenth year. The project will have a useful life of 50 years, counting from the date of the initial investment. The interest rate is 6 %. The cost of operation and maintenance is Rs.200,000 per year. The benefits of the project have been estimated to begin with Rs.1.0 million per year for the first 15 years (at the end of each year), thereafter increasing at once to Rs.2.75 million per year and remaining constant for the remaining 35 years.

**Determine the value of Benefit-cost (B/C) ratio, Net Present Value (B-C), and Internal Rate of Return (IRR).**

#### Question 4

Two alternative routes for a road are being considered. The costs for these alternatives are given in the following table:

|           | Road Construction | Road Maintenance (per year) |
|-----------|-------------------|-----------------------------|
| Project A | Rs. 200 million   | Rs. 15 million              |
| Project B | Rs. 4000 million  | Rs. 8 million               |

If the discount rate is 10% and the project life is 25 years, determine the following:

1. Equivalent Annual Charge (EAC) and NPV of two projects.
2. If only one to be chosen, please suggest the economical choice
3. If we look for social objectives in view, please suggest the economical choice

#### Question 5

In a remote wilderness in Africa a rich ore deposit has been discovered. It has been estimated that all ore can be mined during a period of 20 years. The most economical way to bring out the ore is by river. To make the river navigable there are two alternative projects:

**Plan A** to regulate the river by training works, excavation and blasting of rock, with a total initial cost of **10,000,000** and a cost of dredging of **2,000,000** per year.

**Plan B** to canalize the river by means of weirs and navigation locks: initial costs **20,000,000** and cost of operation and maintenance of **400,000** per year.

Capital for both projects is available at **7.49 %** interest.

The terminal value of the navigation works after 20 years is assumed to be nil.

(A) Make a cost comparison of annual costs.

(B) The cost of dredging of Plan A is now expected to be as follows:

**100,000** during the first year and then gradually increases by an amount of **200,000** per year till it would reach a cost of **3,900,000** during the twentieth year. Determine which of the two projects is more economic.